The ClearSpace-1 'chaser' will be launched into a lower 500km orbit for commissioning and critical tests before being raised to the target orbit for rendezvous and capture using a quartet of robotic arms under ESA supervision. The combined chaser plus Vespa will then be deorbited to burn up in the atmosphere.

According to Dr Holger Krag, head of the Space Safety Program Office at the European Space Agency, the agency is looking into using lasers to gently push the objects off the path. Chris Blackerby, Group COO and Director, Japan for Astroscale said, "I think the idea that space debris is an active threat to our current satellite population is becoming more widely accepted."

NASA scientist Donald Kessler asserted that an exploding chain of space debris can make exploring space and the use of satellites impossible for generations. The fear is that we get to a level of unsustainability of orbit.

An active debris removal mission, if successful, has a positive effect (risk reduction) for all satellites in the same orbital band. This may lead to a dilemma: each stakeholder has an incentive to delay its actions and wait for others to respond. This makes the space debris removal an interesting strategic dilemma. As all actors share the same environment, actions by one have a potential immediate and future impact on all others. This gives rise to a social dilemma in which the benefits of individual investment are shared by all while the costs are not.

To counter this risk, mitigation strategies are now implemented in newly launched satellites.

Following a competitive process, a consortium led by Swiss startup ClearSpace—a spin-off company established by an experienced team of space debris researchers based at Ecole Polytechnique Fédérale de Lausanne (EPFL) research institute—submitted their final proposal.

"This is the right time for such a mission," says Luc Piguet, founder and CEO of ClearSpace. "The space debris issue is more pressing than ever before. Today we have nearly 2000 live satellites in space and more than 3000 failed ones.

"The need is clear for a 'tow truck' to remove failed satellites from this highly trafficked region."

At Space19+, which took place in Seville, Spain, ESA's Ministerial Council agreed to place a service contract with a commercial provider for the safe removal of an inactive ESA-owned object from LEO. Supported within ESA's new Space Safety program, the aim is to contribute actively to cleaning up space, while also demonstrating the technologies needed for debris removal.

Fig. 3: ClearSpace-1 will be the first space mission to remove an item of debris from orbit, planned for launch in 2025. The mission is being procured as a service contract with a startup-led commercial consortium, to help establish a new market for in-orbit servicing and debris removal

"Even if all space launches were halted tomorrow, projections show that the overall orbital debris population will continue to grow, as collisions between items generate fresh debris in a cascade effect," says Luisa Innocenti, heading ESA's Clean Space initiative. "We need to develop technologies to avoid creating new debris and removing the debris already up there."

NASA and ESA studies show that the only way to stabilise the orbital environment is to actively remove large debris items. The ClearSpace-1 mission will target the Vespa (Vega Secondary Payload Adapter) upper stage left in an approximately 800km by 660km altitude orbit after the second flight of ESA's Vega launcher back in 2013. With a mass of 100kg, the Vespa is close in size to a small satellite, while its relatively simple shape and sturdy construction make it a suitable first goal, before progressing to larger, more challenging captures by follow-up missions—eventually including multi-object capture.

Other novel concepts

There are many possible means of reducing the debris hazard to future space operations. There is no shortage of concepts for cleaning up the junk we have left behind in orbit, even if some of them seem far-fetched. Here's an overview of some of the ideas being proposed for cleaning up space debris.

Fig. 4 illustrates a novel concept